Komal Saraf

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EDUCATION:

Bachelor of Technology (B.Tech.) - Computer Science *Sep 2022 - June 2026*
Vellore Institute of Technology (B.TECH CSE Core)
Cumulative GPA - 7.48

SKILLS:

- **Languages**: Python, Java, C++, MySQL.
- **Core Competencies**: Problem-solving, Data Structure and Algorithms, Object-oriented Programming, Database and Management System, Probability, Statistics, Linear Programming, Integer Programming, basic optimization techniques, Machine Learning (TensorFlow, Matplotlib, Pandas, NumPy), Prompt Engineering, version control (Git), CI/CD Pipelines, Model optimization, Data Engineering, Data Visualization, Data Analytics, reporting and trend analysis, data interpretation, technical documentation (product issue logs, user manuals, product release notes and troubleshooting guides), retail demand and fulfilment processes (merchandising buying, product attribution, pricing, promotions, and clearance activities).
- **Tools**: Microsoft Excel, Google Sheets, Google Slides, Tableau, PowerBI, Github.
- **Soft skills**: Analytical, effective oral and written communication, collaborative, teamwork, adaptability.

PROJECTS:

- **CivicHelp AI** *Dec 2025 - Ongoing*
  - Architected and deployed a multimodal GenAI pipeline using Vision Transformers and LLMs to process 10K+ civic complaints, achieving 92% classification F1-score and reducing manual triage time by 60%.
  - Built an embedding + vector search system for similarity detection and hotspot clustering, enabling automated repeated-issue detection and improving routing precision by 45%.
  - Implemented RAG-based policy grounding using municipal SOP embeddings to ensure factually aligned outputs, reducing misinformation risk and improving action recommendation reliability by 30%.
  - Integrated predictive trend and anomaly detection models to forecast rising civic issues and track urgency rates improving efficiency by 35%, develop reliable, scalable, and high-performance software systems.
  - Tech Stack: Python, Generative AI, OpenAI GPT, LangChain, Whisper, YOLO, Vision Transformers, Sentence Transformers, FAISS/Pinecone, Scikit-learn, TensorFlow, FastAPI, Pandas, NumPy, Matplotlib.

- **Heart disease detection** (Machine Learning) *May 2025- Sept 2025*
  - Developed a predictive analytics system for early identification of heart disease risks using patient medical and physical data.
  - Integrated and optimized various machine learning algorithms (e.g., Logistic Regression, Support Vector Machines, Random Forest) to enhance prediction accuracy and efficiency.
  - Attained a prediction accuracy of more than 85% by analyzing 1000+ dataset entries and built visual reports and performance dashboards to communicate insights
  - Skills Used: Python, Scikit-learn, Pandas, NumPy, Matplotlib, ML Model Optimization, Data Analysis.

- **Desktop Assistant** *Nov 2024- Feb 2025*
  - Developed an automated workflow management system to handle routine operational tasks, resulting in a 20% increase in process efficiency.
  - Designed modular workflows to integrate speech recognition for sub-second voice-to-text conversion and a GUI (Tkinter) for task execution features demonstrating hands-on experience in automation, process streamlining, and user-centric design.
  - Skills used: Automation, OpenCV, PyTorch, Tkinter Python libraries (SpeechRecognition, pyttsx3)

CO-CURRICULARS:

- Python gold badge , Hackerrank.
- C++ gold badge , Hackerrank.
- Completed 15 Skill badges in Google Cloud Skill Boost.
- Member, AI Club, VIT Bhopal 2022 – Coordinated and organized multiple events, workshops, and functions within the AI Club, collaborating with cross-functional teams contributing to community building, knowledge sharing, and enhancing student participation in Artificial Intelligence initiatives.